HARD · ADVANCED MATH

Advanced Math

16

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

A model estimates that at the end of each year from 2015 to 2020, the number of squirrels in a population was 150% more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of 2016, there were 180 squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of 2015 and $t \leq 5$?

A. $n = 721.5^t$

B. $n = 722.5^t$

C. $n = 1801.5^t$

D. $n = 1802.5^t$

Q2 ADVANCED MATH

The function f gives the product of a number, x, and a number that is 91 more than x. Which equation defines f?

A. $fx = x^2 + x + 91$

B. $fx = x^2 + 91$

C. $fx = x^2 + 91x$

D. $fx = x^2 + 91x + 91$

Q3 ADVANCED MATH

The surface area of a cube is $6\left(\frac{a}{4}\right)^2$, where a is a positive constant. Which of the following gives the perimeter of one face of the cube?

A. $\frac{a}{4}$

B. a

C. $4a$

D. $6a$

Q4 ADVANCED MATH

When the quadratic function f is graphed in the xy-plane, where $y = fx$, its vertex is -3, 6. One of the x-intercepts of this graph is $-\frac{17}{4}, 0$. What is the other x-intercept of the graph?

A. $-\frac{29}{4}, 0$

B. $-\frac{7}{4}, 0$

C. $\frac{5}{4}, 0$

D. $\frac{17}{4}, 0$

x	gx
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of gx, where $gx = \frac{fx}{x+3}$ and f is a linear function. What is the y-intercept of the graph of $y = fx$ in the xy-plane?

A. $0, 36$

B. $0, 12$

C. $0, 4$

D. $0, -9$

Q6 ADVANCED MATH

For the function q, the value of qx decreases by 45% for every increase in the value of x by 1. If $q0 = 14$, which equation defines q?

A. $qx = 0.5514^x$

B. $qx = 1.4514^x$

C. $qx = 140.55^x$

D. $qx = 141.45^x$

Q7 ADVANCED MATH

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

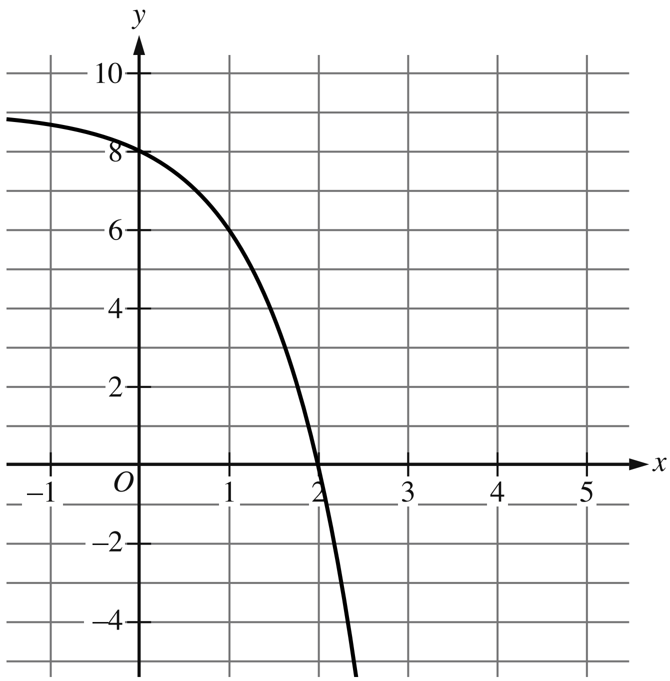
Q8 GRID-IN ADVANCED MATH

For the function f, $f0 = 86$, and for each increase in x by 1, the value of fx decreases by 80%. What is the value of $f2$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
  - In quadrant 2, the curve trends down gradually to the point (0 comma 8).
  - In quadrant 1:
    - The curve trends down gradually to the approximate point (1 comma 6.0).
    - The curve then trends down sharply to the x axis.
  - In quadrant 4, the curve trends down sharply.
- As x decreases, the curve approaches the line y equals 9.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 6)

The graph of $y = fx + 4$ is shown. Which equation defines function f?

A. $fx = \text{-}3^x + 1$

B. $fx = \text{-}3^x + 5$

C. $fx = \text{-}3^x + 8$

D. $fx = \text{-}3^x + 9$

Q10 ADVANCED MATH

For the exponential function f, the value of $f1$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A. $fx = 502^{x+1}$

B. $fx = 802^x$

C. $fx = 1282^{x-1}$

D. $fx = 2052^{x-2}$